

could be taking backups of existing databases or folders and then storing them in Minio. The possibilities are endless, and I recommend readers take a look at the Minio community projects page to see how Minio has been innovatively integrated as part of larger projects.

Minio client SDKs

If you wish to interface with Minio via one or more programming languages, there are SDKs for Java, Python, JavaScript, Go and others. The SDKs are well designed wrappers around the Minio REST APIs. The pattern of using any Minio client SDK is straightforward—you establish a client connection to the Minio server and then perform one or more commands that are applicable. The commands could include listing/creating buckets, listing/uploading file objects, and so on.

A simple backup utility

Let us think of how we can potentially use Minio as a store to retain multiple backups. These could be particular directories that you want to back up, certain files (log files) or even database dumps that you need to save for restore or audit purposes.

To use the API, as mentioned earlier, you will need to have the hostname and port of your Minio server along with the access and secret key. Once you have all this, we can establish a connection to the Minio client and then perform the operation. In this case, we will simply upload a standard file object into a bucket *mybackup* that we have pre-created.

We shall use the Minio Python SDK for this purpose. I have modified the code from the official documentation, and the code assumes that you have a bucket named *mybackup* in the Minio server:

```
from minio import Minio
from minio.error import ResponseError

minioClient = Minio('localhost:9000',
                    access_key='YOUR_MINIO_ACCESS_KEY',
                    secret_key='YOUR_MINIO_SECRET_KEY',
                    secure=False)

print(minioClient)

try:
    minioClient.fput_object('mybackup', 'mydb.log', 'mydb.
log')
except ResponseError as error:
    print(error)
```

The code first creates the Minio client object. Remember to use your access details in the parameters.

It then assumes that you have a *bucketname* created in the Minio server named *mybackup* and a file *mydb.log* in the current directory that you want to upload. The second and third parameters in the *fput_object* method stand for object name (when stored in Minio server) and the file path (where the file can be found by the Python program).

To run the Python app, first ensure that you have downloaded the Minio Python library as given below:

```
pip install minio
```

Minio client

The Minio client (mc) is a command line utility that provides an alternative to UNIX commands like *ls* and *cp*, with the interesting twist that it works for both your local system as well as the Amazon S3 service in a seamless manner. You can check out <https://docs.minio.io/docs/minio-client-complete-guide> for more details.

Contributing to Minio

Minio prides itself on being open source, and a quick look at the GitHub page shows an active project with multiple contributions from developers. The Minio server and client are written using the Go programming language and, hence, contributors need to be familiar with this language. You can also look at the client SDKs available in multiple languages, if you wish to contribute in that way.

You can also contribute to the growing list of Minio community projects that are listed in the community page. A quick glance at how multiple projects have used Minio is sure to inspire you with some ideas.

The creators of Minio have focused on minimalism and providing well designed APIs that make it easy for you to customise it to your needs. This, coupled with the fact that Minio is available on multiple target platforms, allows you to choose your hardware and scale, as needed. 

References

- [1] Minio Home Page: <https://www.minio.io/>
- [2] Minio Downloads: <https://www.minio.io/downloads/#minio-server>
- [3] Minio Client APIs: <https://www.minio.io/downloads/#minio-sdk>
- [4] Minio Blog: <https://blog.minio.io/>
- [5] Minio GitHub Page: <https://github.com/minio/>
- [6] Minio Community Projects: <https://github.com/minio/awesome-minio>

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